

DEFENDED DYNAMIC STOPPING ENVELOPE: REACTION LAG VS. KINETIC BRAKING

Linear Sensitivity to Computational Latency vs. Quadratic Sensitivity to Velocity: $d_{\text{stop}}(v_0, \Delta t) = v_0 \Delta t + v_0^2 / (2a) + \delta_{\text{overhead}} \leq D_{\text{clear}}$

1. LINEAR REACTION TRAVEL

$$d_{\text{react}} = v_0 \cdot \Delta t_{\text{delay}}$$

Unguided coasting during compute latency

2. QUADRATIC BRAKING DISTANCE

$$d_{\text{brake}} = v_0^2 / (2 \cdot a_{\text{brake}})$$

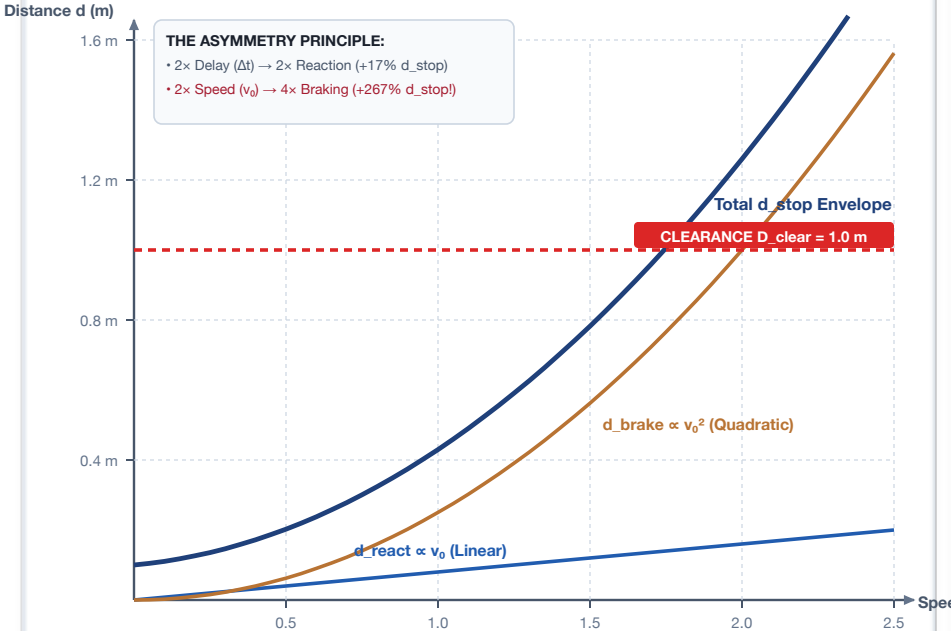
Work-energy dissipation: $E_k = \frac{1}{2} m v_0^2$

3. MAX PERMITTED VELOCITY

$$v_{\text{max}} = -a \Delta t + \sqrt{(a \Delta t)^2 + 2aD_{\text{avail}}}$$

Defended clearance bound: $d_{\text{stop}} \leq D_{\text{clear}}$

ASYMMETRIC PARAMETER SENSITIVITY CURVES



PHYSICAL SCENARIO TRAJECTORIES ($D_{\text{clear}} = 60.0 \text{ cm}$)

CASE 1: Nominal Speed & Latency ($v_0 = 1.0 \text{ m/s}$, $\Delta t = 50 \text{ ms}$)

5cm

Braking 25.0 cm

Margin +20.0 cm ✓

Total Stop $d_{\text{stop}} = 40.0 \text{ cm} \leq 60.0 \text{ cm}$ (CERTIFIED SAFE)

CASE 2: Tail Latency Spike ($v_0 = 1.0 \text{ m/s}$, $\Delta t = 200 \text{ ms}$)

Reaction 20.0 cm (+15cm unguided!)

Braking 25cm

+5cm

Total Stop $d_{\text{stop}} = 55.0 \text{ cm}$ (CLEARANCE MARGIN COMPRESSED TO 5 cm)

CASE 3: Speed Increase + Tail Latency ($v_0 = 1.5 \text{ m/s}$, $\Delta t = 200 \text{ ms}$)

Reaction 30.0 cm

Braking 30cm

Total Stop $d_{\text{stop}} = 86.3 \text{ cm} > 60.0 \text{ cm}$ (CATASTROPHIC IMPACT BREACH!)

FIRST-PRINCIPLES PHYSICAL AI DESIGN LAW

Software models propose speed setpoints; the Body enforces quadratic kinetic momentum. Safe operation requires budgeting P99.9 latency and defending physical clearance.